MATH 151 — Exponential & Logarithmic Functions

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Name: ______________________________ Date: _______________ Score: / 10

Learning Objectives

- Evaluate and expand/condense logarithms using product, quotient, and power rules
- Solve exponential and logarithmic equations
- Apply the change of base formula
- Model exponential growth, decay, and compound interest

Always check for extraneous solutions in log equations — arguments must be positive. For exponential equations without common bases, take ln of both sides.

1. Evaluate the logarithm.

$$\log_2 32$$

Answer: ________________________________

2. Expand using logarithm properties.

$$\log_b\left(\frac{x^3\sqrt{y}}{z^2}\right)$$

Answer: ________________________________

3. Condense into a single logarithm.

$$2\ln x + \ln(x+1) - \ln 5$$

Answer: ________________________________

4. Solve the exponential equation.

$$3^{x+1} = 81$$

Answer: ________________________________

5. Solve the exponential equation using logarithms.

$$5^x = 20$$

Answer: ________________________________

6. Solve the logarithmic equation.

$$\log_3(2x-1) = 4$$

Answer: ________________________________

7. Solve.

$$\ln(x+3)+\ln(x-2)=\ln 14$$

Answer: ______________________________________

8. Use the change of base formula to evaluate.

$$\log_7 50$$

Answer: ______________________________________

9. Compound interest: $5000 invested at 4% compounded continuously. Find the amount after 10 years.

$$A=Pe^{rt},\quad P=5000,\ r=0.04,\ t=10$$

Answer: ______________________________________

10. A population starts at 1000 and doubles every 5 years. Write the exponential model and find the population after 15 years.

$$P(t)=1000\cdot 2^{t/5}$$

Answer: ______________________________________

Problems 2–3: expand vs condense — students confuse these directions. Problem 7 (log equation): extraneous root x=–5 must be rejected. Problems 9–10: applied context.

Solutions

1. Evaluate the logarithm.

$\log_2 32$

→ *log■32 = x means 2^x = 32 = 2■.*
→ *Answer: 5.*

Answer: 5

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2. Expand using logarithm properties.

$\log_b\left(\frac{x^3\sqrt{y}}{z^2}\right)$

→ *log(A/B)=log A–log B. log(AB)=log A+log B. log(A■)=n log A.*
→ *= log x³ + log y^(1/2) – log z² = 3 log x + ½ log y – 2 log z.*

Answer: $3\log_b x + \frac{1}{2}\log_b y - 2\log_b z$

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3. Condense into a single logarithm.

$2\ln x + \ln(x+1) - \ln 5$

→ *2 ln x = ln x². Combine using product/quotient rules.*
→ *ln x² + ln(x+1) – ln 5 = ln[x²(x+1)/5].*

Answer: $\ln\left(\frac{x^2(x+1)}{5}\right)$

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4. Solve the exponential equation.

$3^{x+1} = 81$

→ *81=3■. 3^(x+1)=3■ → x+1=4 → x=3.*

Answer: $x = 3$

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5. Solve the exponential equation using logarithms.

$5^x = 20$

→ *Take ln of both sides: x ln5 = ln20.*
→ *x = ln20/ln5 = log■20 ≈ 1.861.*

Answer: $x = \frac{\ln 20}{\ln 5} \approx 1.861$

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6. Solve the logarithmic equation.

$$\log_3(2x-1)=4$$

→ *Convert to exponential: 3■=2x–1.*
→ *81=2x–1 → 2x=82 → x=41.*

Answer: $x=41$

7. Solve.

$$\ln(x+3)+\ln(x-2)=\ln 14$$

→ *ln[(x+3)(x–2)]=ln14 → (x+3)(x–2)=14.*
→ *x^2+x–6=14 → x^2+x–20=0 → (x+5)(x–4)=0.*
→ *x=4 (reject x=–5 since x–2=–7<0 makes ln undefined).*

Answer: $x=4$

8. Use the change of base formula to evaluate.

$$\log_7 50$$

→ *Change of base: log■50 = ln50/ln7 = log10(50)/log10(7).*
→ *≈ 3.912/1.946 ≈ 2.012.*

Answer: $\dfrac{\ln 50}{\ln 7}\approx 2.012$

9. Compound interest: $5000 invested at 4% compounded continuously. Find the amount after 10 years.

$$A=Pe^{rt},\quad P=5000,\ r=0.04,\ t=10$$

→ *A = 5000·e^(0.04·10) = 5000·e^0.4.*
→ *e^0.4 ≈ 1.4918. A ≈ 5000·1.4918 ≈ $7459.*

Answer: $A=5000e^{0.4}\approx \$7459$

10. A population starts at 1000 and doubles every 5 years. Write the exponential model and find the population after 15 years.

$$P(t)=1000\cdot 2^{t/5}$$

→ *Doubling time 5 yrs → P(t)=1000·2^(t/5).*
→ *P(15)=1000·2^(15/5)=1000·2^3=1000·8=8000.*

Answer: $P(15)=1000\cdot 2^3=8000$